Find an initial solution by using north west corner rule. What is the total cost for this solution?

2. Consider the following transportation problem

	Destination				Availability
	D_1	D_2	D_3	D_4	
O_1	5	8	3	6	30
O_2	4	5	7	4	50
O_3	6	2	4	6	20
Requirement	30	40	20	10	

Determine an initial basic feasible solution using (a) Least cost method (b) Vogel's approximation method.

3. Determine an initial basic feasible solution to the following transportation problem by using (i)North West Corner rule (ii) least cost method.

		Destination			Supply
		D_1	D_2	D_3	
Source	S_1	9	8	5	25
	S_2	6	8	4	35
	S_3	7	6	9	40
	Requirement	30	25	45	

4. Explain Vogel's approximation method by obtaining initial basic feasible solution of the following transportation problem.

		Destination				Supply
		D_1	D_2	D_3	D_4	
Origin	O_1	2	3	11	7	6
	O_2	1	0	6	1	1
	O_3	5	8	15	9	10
	Demand	7	5	3	2	

5. A car hire company has one car at each of five depots a,b,c,d and e. A customer in each of the fine towers A,B,C,D and E requires a car. The distance (in miles) between the depots (origins) and the towers(destinations) where the customers are given in the following distance matrix.

	a	b	c	d	e
A	160	130	175	190	200
B	135	120	130	160	175
C	140	110	155	170	185
D	50	50	80	80	110
E	55	35	70	80	105

How should the cars be assigned to the customers so as to minimize the distance travelled?

6. A natural truck-rental service has a surplus of one truck in each of the cities 1,2,3,4,5 and 6 and a deficit of one truck in each of the cities 7,8,9,10,11 and 12. The distance(in kilometers) between the cities with a surplus and the cities with a deficit are displayed below:

		To					
		7	8	9	10	11	12
From	1	31	62	29	42	15	41
	2	12	19	39	55	71	40
	3	17	29	50	41	22	22
	4	35	40	38	42	27	33
	5	19	30	29	16	20	33
	6	72	30	30	50	41	20

How should the truck be dispersed so as to minimize the total distance travelled?

7. A person wants to invest in one of three alternative investment plans: Stock, Bonds and Debentures. It is assumed that the person wishes to invest all of the funds in a plan. The pay-off matrix based on three potential economic conditions is given in the following table:

XII Std - Business Maths & Stat EM Chapter 10.indd 262 2/29/2020 11:45:22 AM